

Program 6

Docker Image Build, Push to Docker Hub & GitHub Container Registry

1. Create project directory

```
mkdir program-6 && cd program-6
```

Step 1: Create a Dockerfile

```
nano Dockerfile
```

```
FROM alpine:latest
```

```
# a simple command
```

```
CMD ["echo", "Hello Srusti, this is a small image!"]
```

Step 2: Build the Docker Image

```
docker build -t program-6 .
```

Step 3: Tag the Image for Docker Hub

```
docker tag program-6 <dockerhub-username>/program-6:latest
```

Step 4: Push the Image to Docker Hub

Login to Docker Hub using your Docker Hub credentials (Personal Access Token as password).

```
docker login -u <dockerhub-username>
```

Push the image to Docker Hub:

```
docker push <dockerhub-username>/program-6:latest
```

Step 5: Tag the Image for GitHub Container Registry (GHCR)

Tag the same image for GitHub Container Registry.

```
docker tag program-6 ghcr.io/github-username/program-6:latest
```

Step 6: Push the Image to GHCR

Login to GHCR using GitHub Personal Access Token (PAT).

```
echo $GITHUB_PAT | docker login ghcr.io -u <github-username>  
--password-stdin
```

Push the image:

```
docker push ghcr.io/github-username/program-6:latest
```

Step 7: Connect the Local Project to GitHub (Source Code)

```
git init
```

```
git remote add origin https://github.com/<github-username>/program-6.git
```

```
git remote set-url origin
```

```
https://github.com/<github-username>/program-6.git
```

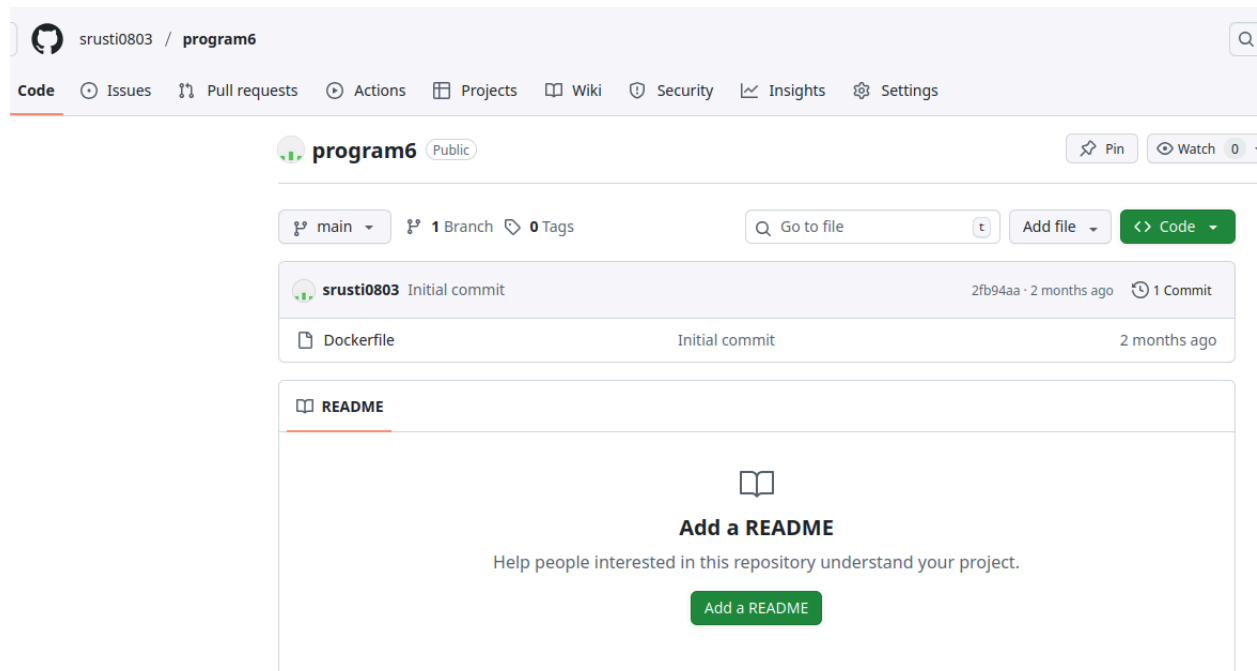
```
git remote -v
```

Step 8: Push the Source Code to GitHub

```
git add .
```

```
git commit -m "Initial commit"
```

```
git push -u origin main
```



Step 9: Verify Uploads on Both Registries GitHub

- Ensure the repository program-6 exists on GitHub.
- Navigate to Packages.

Docker Hub

- Login to Docker Hub.
- Go to Repositories.

The screenshot shows the Docker Hub interface for the repository `srusti0803/srusti`. The page is titled "Repositories / srusti / General" and shows the repository details, including the last push time and repository size. The "Tags" section shows a table with one tag, `latest`, which was pushed about 6 hours ago. The "Repository overview" section is also visible, along with a sidebar on the left containing navigation links like "Repositories", "Hardened Images", "Collaborations", "Settings", "Billing", "Usage", "Pulls", and "Storage".

login.docker.com/device/ x The URL you requested has l x login.docker.com/activat x SRUSTI GC | Docker Home x srusti0803/srusti general x +

hub.docker.com/repository/docker/srusti0803/srusti/general

hub Explore My Hub Search Docker Hub CtrlK Sign in

srusti0803 Docker Personal

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories / srusti / General

srusti0803/srusti

Last pushed about 6 hours ago · Repository size: 137.6 MB · ☆0 · ⬇0

Add a description

Add a category

General Tags Image Management BETA Collaborators Webhooks Settings

Tags

This repository contains 0 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	about 6 hours

See all

Repository overview

An overview describes what your image does and how to run it.

Add overview

DOCKER SCOUT INACTIVE Activate

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

Go to Docker Build Cloud →

Using 1 of 1 private repositories.

Docker commands

To push a new tag to this repository:

```
docker push srusti0803/srusti:tagname
```